

MALUNG A, Sweden

WMO: 024070

Lat: 60.6765N

Lon: 13.7072E

Elev: 306

StdP: 97.70

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.1	-22.1	-27.6	0.3	-25.0	-24.4	0.4	-22.0	9.5	0.0	7.6	0.7	0.6	320	0.627

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	25.7	16.1	23.7	15.4	21.8	14.6	17.6	22.8	16.6	21.4	15.7	20.3	2.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.9	11.7	18.4	14.8	10.9	18.1	13.7	10.2	17.1	50.7	22.9	47.7	21.4	44.9	20.4	22.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.3	6.2	5.3	DB	-26.6	27.7	3.4	2.1	-29.0	29.2	-31.0	30.4	-32.8	31.6	-35.3	33.1	(4)
(5)			WB	-26.4	18.9	3.1	1.7	-28.6	20.1	-30.4	21.0	-32.2	22.0	-34.4	23.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.6	-6.9	-6.2	-2.4	3.1	8.0	12.2	15.2	13.4	9.2	3.7	-1.0	-5.5	(6)	
	DBStd	8.93	6.65	6.18	5.00	3.45	3.88	3.05	2.84	2.82	3.23	3.93	4.82	6.96	(7)	
	HDD10.0	2732	523	453	384	209	86	12	1	4	52	197	332	479	(8)	
	HDD18.3	5379	782	686	642	458	319	186	106	153	275	453	582	738	(9)	
	CDD10.0	404	0	0	0	1	25	77	161	112	27	2	0	0	(10)	
	CDD18.3	9	0	0	0	0	0	1	7	1	0	0	0	0	(11)	
	CDH23.3	199	0	0	0	0	6	31	131	32	0	0	0	0	(12)	
	CDH26.7	21	0	0	0	0	0	2	18	1	0	0	0	0	(13)	
(14) Wind	WSAvg	2.2	1.9	2.1	2.3	2.3	2.4	2.4	2.2	2.0	2.0	2.1	2.1	2.1	(14)	
(15) Precipitation	PrecAvg	703	44	32	34	44	54	74	88	76	75	68	63	50	(15)	
	PrecMax	1080	98	75	68	123	172	176	195	153	161	175	154	134	(16)	
	PrecMin	501	8	6	1	1	11	10	28	13	17	10	8	8	(17)	
	PrecStd	103	24	16	19	29	31	38	37	39	35	39	30	27	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.2	6.2	11.9	18.4	24.3	26.3	28.7	26.1	20.8	14.6	10.1	7.8	(19)	
		MCWB	3.7	3.4	6.1	9.8	14.0	15.5	17.7	16.3	14.7	11.5	8.9	6.5	(20)	
	2%	DB	3.7	4.7	8.4	15.3	21.8	23.8	26.7	24.1	18.4	12.5	8.7	5.9	(21)	
		MCWB	2.6	2.8	4.0	8.3	12.9	14.8	17.0	16.1	13.1	10.4	7.4	4.8	(22)	
	5%	DB	2.7	3.3	6.2	12.6	19.4	21.4	24.6	22.0	16.5	11.2	6.9	4.4	(23)	
		MCWB	1.8	1.8	2.7	6.9	11.8	13.9	16.1	15.1	12.5	9.5	5.9	3.4	(24)	
	10%	DB	1.4	2.0	4.5	10.2	16.5	19.2	22.5	19.9	14.9	10.0	5.5	2.9	(25)	
		MCWB	0.7	0.7	1.9	5.5	10.3	12.7	15.2	14.5	11.7	8.5	4.7	2.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.3	4.6	6.5	10.7	15.5	17.5	19.1	18.2	15.8	12.4	9.2	6.8	(27)	
		MCDB	4.8	5.5	10.7	16.5	21.0	22.8	25.1	22.4	18.6	13.7	9.9	7.6	(28)	
	2%	WB	2.9	3.1	4.6	8.9	14.0	16.1	18.1	17.1	14.3	11.1	7.5	5.0	(29)	
		MCDB	3.5	4.3	7.5	14.1	20.0	21.5	24.1	20.7	16.8	12.1	8.4	5.7	(30)	
	5%	WB	1.9	2.0	3.4	7.5	12.4	14.7	17.1	16.2	13.3	9.9	6.0	3.5	(31)	
		MCDB	2.5	3.1	5.6	11.7	17.9	20.2	22.8	20.3	15.6	10.9	6.7	4.1	(32)	
	10%	WB	0.8	0.9	2.2	6.1	11.0	13.5	16.1	15.3	12.2	8.8	4.8	2.1	(33)	
		MCDB	1.3	1.7	4.3	9.4	16.3	18.0	21.0	19.2	14.3	9.9	5.4	2.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.0	8.7	9.9	10.3	11.4	10.8	11.4	10.7	9.6	7.5	5.7	6.7	(35)	
		MCDBR	5.7	8.2	11.4	15.4	17.0	15.2	15.7	14.8	12.4	8.5	6.5	5.7	(36)	
	5% WB	MCWBR	5.4	6.6	7.5	9.1	9.0	7.5	7.6	7.6	7.7	6.6	6.0	5.4	(37)	
		MCWBR	5.6	6.9	9.9	13.8	14.2	12.9	13.2	12.0	10.1	6.7	5.9	5.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.238	0.284	0.311	0.349	0.351	0.344	0.363	0.354	0.323	0.309	0.260	N/A	(40)	
		taud	2.187	2.285	2.334	2.372	2.434	2.478	2.437	2.456	2.532	2.480	2.267	N/A	(41)	
	Ebn at Noon		533	715	817	849	874	886	857	836	808	690	499	N/A	(42)	
		Edh at Noon	37	64	85	99	101	98	100	91	71	54	35	N/A	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.28	0.94	2.45	3.88	4.71	5.34	4.79	3.75	2.48	1.11	0.35	0.17	(44)	
	RadStd		0.04	0.18	0.27	0.46	0.47	0.36	0.65	0.39	0.31	0.21	0.10	0.02	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (48 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page